TreeLoom: Whole-Tree Learning over Ordered Genealogies for Local Ancestry Inference

Anonymous Authors

Abstract

A person's genome contains segments inherited from different ancestral populations. Identifying these segments helps reconstruct population history, but their boundaries can be difficult to recover from genetic data. Inferred genealogies offer a natural representation of shared ancestry: each local tree connects sampled genomes, and the trees change along the chromosome. Learning from this representation requires combining evidence within each tree with continuity across trees. We develop TreeLoom, a whole-tree neural model that propagates genetic and reference-panel information through complete genealogies and follows sampled haplotypes along the genome. Structural conditioning marks changes in descendant groups, while recurrent context connects local predictions. On eight European development simulations, TreeLoom reaches $97.08 \pm 0.57\%$, compared with $94.45 \pm 1.15\%$ for the evaluated ARGMix release. Matched controls show that much of the improvement within our model family also follows from continued training. The highest span accuracy and the best boundary score occur in different variants. These findings support whole-tree learning as a promising ancestry framework and identify boundary recovery as a distinct target for model design.

1 Introduction

Local ancestry inference identifies the ancestral source of each segment of a genome. It can reveal how populations mixed, how ancestry changed through time and how genetic variants moved between populations. In ancient genomics, dated samples provide observations from different stages of this history. A useful predictor must distinguish reference ancestries while respecting both the genealogical relationships between samples and the spatial continuity of ancestry along a chromosome.

An ancestral recombination graph (ARG) represents the changing genealogy of sampled genomes. At any genomic position, a marginal tree connects the sampled haplotypes through common ancestors. Recombination changes this tree along the chromosome. The resulting data have two linked forms of structure: relationships within a tree and correspondence between successive trees. Sampled haplotypes retain their identities, even when their inferred ancestral relationships change.

Recent ancestry methods exploit these genealogical representations. AncestralPaths uses ancestral-path features in the study of ancient Eurasian genomes (Irving-Pease et al. 2024). ARGMix combines genealogical graph representations with genomic sequence modeling (Shanks et al. 2026). ARGformer learns representations of genealogical paths (Bonet et al. 2026). These approaches motivate a complementary question: how much ancestry information can a compact model recover by processing complete marginal trees and maintaining a continuous sampled-haplotype representation?

A generic sequence of graph embeddings does not directly preserve every sample's relationship to all other samples. Conversely, a set-based graph model does not by itself encode genomic order or physical distance. For genealogies, treating internal-node identifiers as persistent can also introduce a false correspondence. The reliable shared interface is the sampled haplotype. Descendant sets provide an additional structural comparison, but an equal descendant set does not prove that two inferred internal nodes represent the same biological ancestor.

We therefore build TreeLoom around two complementary computations. Upward and downward messages connect reference and query haplotypes through every node in each local genealogy. A genomic recurrent network then follows each sampled haplotype through the ordered tree sequence. Local changes in descendant groups condition a second tree traversal. We examine whether additional cross-tree context improves this representation using matched episode and memory controls.

The present results support three scientific observations. First, the evaluated whole-tree pipelines exceed the evaluated native ARGMix and AncestralPaths pipelines on the shared European development endpoint. Second, simple local structural features are competitive with the examined cross-tree interactions. Third, span accuracy and tract-boundary recovery favor different variants. This analysis identifies which design choices improve fitted ancestry prediction and which metrics expose remaining errors. The completed comparisons use eight development simulations that have also informed model selection. Independent testing and the remaining native benchmarks are explicit completion items in this working manuscript.

2 Related Work

Genealogy-based ancestry inference. AncestralPaths accompanies an analysis of the selection landscape and genetic legacy of ancient Eurasians (Irving-Pease et al. 2024). Its ancestral-path formulation distinguishes historical routes that can share the same broad source ancestry. This distinction matters when aligning its output with a four-source classifier. Our comparison uses the released dense classifier and collapses its six path probabilities to four source probabilities. The original study describes a convolutional classifier, whereas the archived executable model is dense. We label the comparison by the executed archive.

ARGMix learns ancestry from genealogical graph representations and genomic context (Shanks et al. 2026). We evaluate a current author release with its native model and selection rule, using a disclosed panel adaptation. ARGformer instead learns genealogical path representations with a transformer and supports ancestry assignment by retrieval (Bonet et al. 2026). Its American admixture benchmark motivates our separate three-source comparison. A completed ARGformer head-to-head score is not yet available. FLARE uses a haplotype-based local ancestry framework (Browning, Waples, and Browning 2023) and provides a comparator that does not require the same neural genealogy representation.

These methods differ in what they encode, how they obtain supervision and how they select predictions. A shared observed panel and a common scorer make their ancestry outputs comparable on that endpoint. They do not make the comparison a controlled test of architecture alone.

Learning from collections of graphs. Graph Set Transformer interleaves node-level propagation and set-wide contextual modeling rather than first reducing each graph to a fixed embedding (Escrig Molina, Chen, and Probst 2026). This provides a relevant motivation for exchanging information before final graph readout. Our setting adds genomic order, unequal interval lengths and persistent sampled entities. These properties determine which cross-tree correspondences have a biological interpretation.

Temporal graph state-space models study recurrent state on changing graphs (Li et al. 2024). Gated recurrent units provide an established sequence mechanism (Cho et al. 2014). Our graph sweeps, gated recurrence, pooling and residual conditioning use established ingredients. The empirical question is whether their genealogical organization improves ancestry inference, and whether a proposed cross-tree interaction adds value beyond matched local and continued-training controls.

3 TreeLoom

3.1 Model Family and Variants

TreeLoom names our full model with memory along persistent sampled haplotypes. Every family member combines a whole-tree encoder with a genomic recurrent network. The suffix identifies the changed component: **Base** has neither structural residuals nor added sample memory. **Clade** adds conditioning on changes in descendant groups. **Local** also transforms a node's own embedding, whereas **Episode** replaces that local residual with information from other trees in a contiguous clade episode. **NoMem** extends Local with the memory comparison's training budget. **Group** and **Pair** replace stable-sample memory with cut-group and pair-cell transport, respectively. Figure 1 depicts the full TreeLoom model. The MLP and Tensor suffixes denote alternative clade-event residuals, described in the supplement. The Base–Clade–Local progression also increases training exposure and is not a set of matched-budget ablations. The final four-arm memory comparison does use a common initialization and run budget.

3.2 Inputs and Prediction Target

Let $T_t = (V_t, E_t)$ be the inferred marginal tree on interval I_t of length w_t. The same sampled haplotypes appear in every tree. Let $v_t(q)$ denote the node of sampled haplotype q in tree t. Inferred internal nodes need not correspond across trees.

The observed inputs are phased genetic variants, inferred topology and branch ages, sample ages, and source labels for the reference panel. The source labels of query haplotypes are masked. The model predicts a probability vector p_{tq} over Anatolian farmer (Ana), Caucasus hunter-gatherer (CHG), western hunter-gatherer (WHG) and eastern hunter-gatherer (EHG) ancestry. Training uses simulated ancestry labels as external supervision. These labels do not define input features, inferred trees or genomic intervals.

Figure 1 shows the data representation and the strongest span-accuracy variant. The central design preserves the complete genealogy during graph propagation and the complete sequence of marginal-tree contexts during genomic recurrence.

3.3 Genetic Evidence and Whole-Tree Propagation

The genetic encoder summarizes each haplotype's called alleles relative to observed donor and panel allele frequencies. Mean and maximum pooling within a tree interval produce a local genetic representation. This representation is combined with sample age and the known reference label, where available. Inferred node ages and node roles provide features for internal ancestors. These quantities produce an initial node state b_v. Branch geometry and mutation-concordance summaries determine message gates.

The first traversal propagates information from sampled leaves toward the root. For parent v and child u,

$$m_{u\to v} = \frac{g_u^{\uparrow} \odot W_{\uparrow} h_u}{\max(1, d_v)}, \tag{1}$$

$$s_v = \sum_{u \in \mathrm{child}(v)} m_{u\to v}, \tag{2}$$

$$h_v = \mathrm{LN}(b_v + \tanh(A_{\uparrow}[b_v, s_v] + a_{\uparrow})), \tag{3}$$

where d_v is the number of children, $g_u^{\uparrow}$ is an observed-feature gate and brackets denote concatenation.

The downward traversal supplies the context outside each child's branch. Its parent context excludes the recipient's own upward message,

$$c_{v\setminus u} = o_v + b_v + s_v - m_{u\to v}. \tag{4}$$

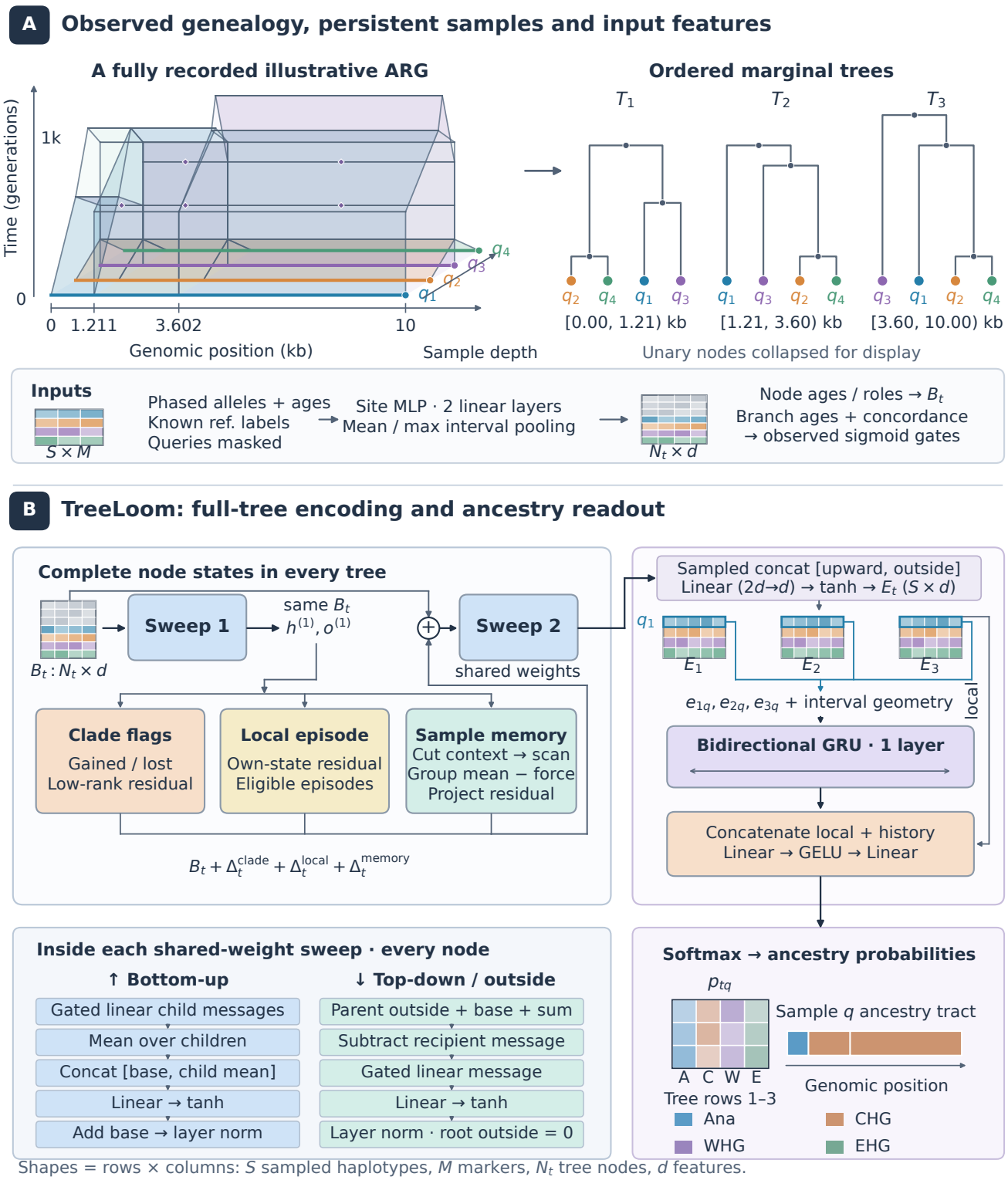


Figure 1: TreeLoom architecture. **A:** An illustrative ARG with recorded recombination events yields three marginal trees. Sheets follow genomic edges. Tip colors identify samples. **B:** Two shared whole-tree sweeps, local residuals, sample memory and a one-layer bidirectional GRU feed the ancestry classifier. Outputs are schematic.

A gated learned transformation maps this context to the child's outside state o_u. Root outside states are zero. Excluding the recipient contribution prevents its immediate return as outside evidence. A learned projection of the sampled node's upward and outside states gives its local tree representation. Both traversals visit every native node.

3.4 Structural Conditioning

For a node v, let $C_t(v)$ be the set of its descendant sampled haplotypes. We compare proper clades between adjacent trees. A proper clade contains at least two sampled haplotypes and excludes the full panel. Equal clades along a unary chain use the deepest structural representative.

A node receives a gained flag g_v when its clade is absent from the preceding tree and a lost flag ℓ_v when it is absent from the following tree. End-of-sequence missing-neighbor flags are zero. These flags describe observed topology changes rather than ancestral identity. They condition a local residual,

$$\Delta_v^{\text{clade}} = O[g_v(Uh_v \odot Vh_v) - \ell_v(Uh_v \odot Wh_v)]\,. \quad (5)$$

The projections use a low-rank intermediate representation. A second whole-tree sweep receives $b_v + \Delta_v$ and reuses the first sweep's learned weights and observed gates.

We also examine contiguous episodes of the same proper clade. For episode member i, a span-weighted context excludes its own tree,

$$\bar{h}_{-i} = \frac{\sum_{j\in E(i)} w_j h_j - w_i h_i}{\sum_{j\in E(i)} w_j - w_i}. \quad (6)$$

Episodes with at least two members receive the residual

$$\Delta_i^{\text{episode}} = O_e\big[(U_e h_i) \odot (V_e \bar{h}_{-i}) + W_e h_i\big]\,. \quad (7)$$

The matched local control substitutes h_i for $\bar{h}_{-i}$ on the same eligible nodes. It retains the same parameter count, initialization and training budget. This control asks whether foreign-tree states help beyond the additional local transformation.

3.5 Memory along Persistent Samples

TreeLoom adds stable-sample memory to TreeLoom-Local. TreeLoom-NoMem retains the same local architecture without this added memory. A time cut at 350 generations groups sampled haplotypes by their observed ancestor at that cut. A sample older than the cut retains its sampled node. This cut is computed from inferred node ages, without ancestry labels.

For cut group k in tree t, a learned projection of upward and outside context gives a forcing vector f_{tk}. Let $k_t(q)$ denote haplotype q's group. A forward memory follows the same sampled haplotype across trees,

$$z^{\rightarrow}_{tq} = \rho_t z^{\rightarrow}_{t-1,q} + (1-\rho_t) f_{t,k_t(q)}, \qquad \rho_t = \exp(-w_t/\lambda). \quad (8)$$

The first state is initialized by its forcing. A reverse scan uses the corresponding reverse order. The positive learned distance scale λ lets physical span determine the amount of retained history.

The two directions are averaged, pooled over the current cut group, and centered by subtracting the current forcing. A learned output projection injects this residual at that group's representative before the second tree sweep. This mechanism preserves sample identity even when cut groups change. TreeLoom-Pair uses pair-cell memory, which transports history through intersections of adjacent cut groups, while TreeLoom-Group discards the additional intersection conditioning. TreeLoom-NoMem supplies the matched-budget control without added memory.

3.6 Genomic Context and Ancestry Readout

The second tree sweep produces a local representation e_{tq} for every sampled haplotype. A bidirectional recurrent network receives this representation together with interval length and neighboring interval distances,

$$r_{1:T,q} = \text{BiGRU}\big([e_{tq}, \text{geometry}(I_t)]_{t=1}^{T}\big). \quad (9)$$

The classifier uses both local and genomic information,

$$p_{tq} = \text{softmax}\left(\text{MLP}([e_{tq}, r_{tq}])\right). \quad (10)$$

The recurrent computation retains all tree intervals and all sampled haplotypes. Query selection occurs only at readout. Predictions are constant within each native tree interval, which limits the possible positions of ancestry transitions.

4 Experimental Design

4.1 European Ancestry Panel

The shared European development panel contains 476 diploid individuals, comprising 952 phased haplotypes. Of these, 266 are reference haplotypes and 686 are queries. The query cohorts include present-day individuals, Bronze Age individuals, Bronze Age Anatolians (BAA), Neolithic farmers (Neo) and Yamnaya (Yam). The released cohort names are retained. Bronze Age is an approximate correspondence to the preprint's Post-Neolithic category, rather than a verified identical panel.

We generated 32 training and eight development-validation simulations, each spanning 2 Mb, from the released ancient-European demographic formulation. Relate inferred the genealogies from observed variants (Speidel et al. 2019). The simulation and inference parameters are given in the supplement, including demographic admixture events and sample-panel sizes. Simulation identity determines the split before any rows, tree intervals or sequence chunks are constructed.

The four-source endpoint uses Ana, CHG, WHG and EHG. AncestralPaths' six path probabilities are collapsed before class assignment,

$$(p_{\text{Ana}}, p_{\text{CHG}}, p_{\text{WHG}}, p_{\text{EHG}}) = (p_1 + p_5, p_2 + p_6, p_3, p_4). \quad (11)$$

The native six-path task is distinct from this derived four-source endpoint.

4.2 Metrics and Statistical Unit

All European pipelines are scored on the same fixed 100 bp bins, clipped to inferred coverage. Ancestry truth is intersected with these bins to obtain the true physical mass m_{iqc}

of source c for query q in bin i. Span accuracy is

$$A = \frac{\sum_{iqc} m_{iqc} \mathbf{1}\{c = \arg\max_k p_{iqk}\}}{\sum_{iqc} m_{iqc}}. \quad (12)$$

The primary endpoint gives each simulation equal weight and considers the 182 present-day query haplotypes. Cohort and all-query results are secondary. Negative log-likelihood (NLL) weights true-class log probabilities by the same physical mass.

Span accuracy emphasizes genomic extent. Boundary precision, recall and F1 instead evaluate transitions using one-to-one positional matching within 1 kb. This matcher ignores source-transition identity. We report mean and sample SD across the eight simulations. These SDs quantify variation among simulations for one fitted model per condition, rather than variability across training seeds or a confidence interval.

4.3 Native Comparators and Matched Controls

AncestralPaths retains the released dense classifier, six outputs and five-epoch fit. Its bounded sampler has a shortfall in BAA path-six rows, as detailed in the supplement. ARGMix retains the current author's released model, its 15-reference representation without ancestor nodes, genomic sequence modeling and native early stopping. A full-panel slot and metadata adaptation lets it process the common observed roster. The preprint instead specifies 25 references and three ancestor nodes, and the public release does not support ancestor inputs. We label the evaluated release condition explicitly. The model's inner selection rule remains unchanged.

Our model also receives observed SNP features. The native pipelines differ in representations, row sampling, supervision units, training exposure and checkpoint selection. The common endpoint therefore compares fitted ancestry pipelines. It does not assign their entire difference to the neural architecture.

Within our model family, matched controls isolate narrower questions. Clade-event tensor and event-MLP models share added capacity and training. Local flags use the same added parameter count and continuation budget. Episode models differ only in foreign versus own-state conditioning. The final four-arm memory comparison starts from the same local-control checkpoint and gives every arm the same additional training budget. All model weights remain trainable. Training details and cumulative exposure are in the supplement.

Research TODO. Complete the untouched European assessment and common-data native ARGformer comparison. For a comparison with the ARGMix preprint's numerical tables, verify the original sample roster, reference eligibility, genomic map, observation pipeline and scoring protocol. The available author release postdates the preprint and does not support its ancestor inputs. Recover the original paper-era model, training specification and checkpoint. A 25-reference release sensitivity would still have zero ancestors and must be named separately. The completed 476-diploid condition remains a shared development comparison.

Pipeline	Span accuracy (%) ↑
Released AncestralPaths, derived C4	93.78 ± 0.84
Current native ARGMix	94.45 ± 1.15
TreeLoom-Base	95.49 ± 0.83
TreeLoom-Clade	96.58 ± 0.60
TreeLoom-Local	96.82 ± 0.61
TreeLoom-Episode	96.83 ± 0.59
TreeLoom-NoMem	97.00 ± 0.50
TreeLoom	$\mathbf{97.08 \pm 0.57}$

Table 1: Shared European development endpoint on 182 present-day query haplotypes. Values are mean±simulation SD across eight 2 Mb simulations. The best observed mean is bold. Inputs, training exposure and model selection differ between pipeline families. This is not a reproduction of published table values.

5 Results

5.1 Shared European Development Performance

Table 1 summarizes the evaluated pipelines. TreeLoom reaches $97.0778 \pm 0.5674\%$ present-day span accuracy, compared with $94.4461 \pm 1.1491\%$ for current native ARGMix and $93.7830 \pm 0.8385\%$ for the derived AncestralPaths output. The TreeLoom–ARGMix difference is 2.63 percentage points for these fitted pipelines.

The progression within the neural family also increases training exposure. TreeLoom-NoMem reaches 97.00% without adding memory. TreeLoom exceeds TreeLoom-NoMem by only 0.0778 points, and the paired episode comparison adds only 0.0096 points for foreign states. Thus the larger pipeline gain does not establish that cross-tree message exchange is its cause.

The source cohorts have different prediction difficulty (Table 2). TreeLoom exceeds the evaluated native comparators in every listed cohort under this condition. Scores are highest for the ancient aggregate. This does not demonstrate equal performance on empirical ancient DNA, since source mixtures, sampling ages and tract structures differ between cohorts.

5.2 What Does Cross-Tree Context Add?

TreeLoom-Clade outperforms the examined TreeLoom-Tensor interaction by 0.3104 ± 0.2588 accuracy points and the matched TreeLoom-MLP interaction by 0.3771 ± 0.2165 points. The simpler structural control therefore captures useful information that these interactions need not exchange explicitly. In a later matched comparison, TreeLoom-Episode exceeds TreeLoom-Local by 0.0096 ± 0.1527 points. This falls below the registered practical-gain threshold of 0.10 points.

The ordered-memory comparison tests whether a richer transport rule improves over stable samples and continued training (Table 3). TreeLoom-Pair exceeds TreeLoom-Group by 0.0137 ± 0.0243 accuracy points. It is below TreeLoom by 0.0138 ± 0.0384 points. Neither registered utility criterion is met: a gain of at least 0.10 accuracy points without a boundary-F1 decrease, or a gain of at least 1.00 boundary-F1 point with no more than 0.05 accuracy points lost, against

Cohort	Query haplotypes	AncestralPaths, C4	ARGMix	TreeLoom-NoMem	TreeLoom
Present-day	182	93.78 ± 0.84	94.45 ± 1.15	97.00 ± 0.50	$\mathbf{97.08 \pm 0.57}$
Bronze Age	162	94.76 ± 1.18	97.71 ± 0.75	99.14 ± 0.40	$\mathbf{99.16 \pm 0.43}$
BAA	14	96.94 ± 1.32	98.44 ± 1.02	99.25 ± 1.04	$\mathbf{99.35 \pm 1.04}$
Neo	302	97.57 ± 0.27	99.30 ± 0.26	99.65 ± 0.06	$\mathbf{99.73 \pm 0.07}$
Yam	26	95.43 ± 1.41	99.22 ± 0.90	99.91 ± 0.14	$\mathbf{99.94 \pm 0.12}$
Ancient	504	96.54 ± 0.32	98.76 ± 0.31	99.49 ± 0.13	$\mathbf{99.55 \pm 0.13}$
All queries	686	95.81 ± 0.45	97.61 ± 0.50	98.83 ± 0.22	$\mathbf{98.89 \pm 0.23}$

Table 2: Cohort span accuracy (%), mean±simulation SD across the same eight development simulations. TreeLoom is the full stable-sample-memory model. TreeLoom-NoMem is its matched-budget continuation without added memory. Bold identifies the best mean within each row. Ancient and all-query rows aggregate the cohorts above them. Bronze Age has not been established as identical to the published Post-Neolithic panel.

TreeLoom variant	Accuracy ↑	Boundary F1 ↑
TreeLoom-Pair	97.06 ± 0.56	18.93 ± 1.42
TreeLoom-Group	97.05 ± 0.58	18.85 ± 1.05
TreeLoom	$\mathbf{97.08 \pm 0.57}$	18.38 ± 1.31
TreeLoom-NoMem	97.00 ± 0.50	$\mathbf{19.05 \pm 1.43}$

Table 3: Matched memory comparison, mean±simulation SD in percent. Pair and Group substitute the memory transport rule. NoMem omits added sample memory. All arms finish the same additional training budget, with checkpoint selection individual to each arm. Bold identifies the largest mean per metric. Boundary tolerance is 1 kb.

every control. These thresholds are practical development criteria.

TreeLoom has the highest span accuracy, while TreeLoom-NoMem has the highest boundary F1. Their paired differences are $+0.0778$ accuracy points and -0.6682 boundary-F1 points. This tradeoff is central to evaluating a method intended to recover ancestry tracts.

5.3 Tract Boundaries Remain Difficult

Table 4 shows that improvements in span accuracy can accompany better boundary F1 relative to native pipelines, while still leaving most true transitions unrecovered. TreeLoom reaches $18.38 \pm 1.31\%$ F1, while TreeLoom-NoMem reaches $19.05 \pm 1.43\%$. AncestralPaths has the highest recall in this comparison, at a much lower precision.

A diagnostic of frozen TreeLoom-Base predictions separates representation from recognition. Of 4,008 true changes, 1,094 have no native interval edge within 1 kb. Yet 2,195 of the model's 3,289 missed changes do have a nearby candidate. Finer readout alone would therefore address only part of the problem. Short false ancestry islands also contribute disproportionately to extra boundaries: islands of at most 10 kb account for $35.46 \pm 2.71\%$ of extra boundaries but only $3.83 \pm 0.73\%$ of erroneous physical mass. This explains why a span-based ranking can understate fragmentation.

5.4 American Admixture Benchmark

A separate 30 Mb American simulation contains 600 reference and 1,000 query haplotypes with EUR, EAS and AFR source classes. Released FLARE calls reach 98.0100% accuracy and 97.7151% macro F1 on 10,000 predetermined focal records. This is one validation simulation, so no between-simulation SD is available. The endpoint counts focal records rather than physical span.

Research TODO. Complete our model and native ARGformer on the same American observations, reference/query identities and focal support. Add multi-simulation uncertainty and verify the published demographic and prediction settings before comparing with its Table 1. The current FLARE score alone establishes no head-to-head ranking.

5.5 Application to Ancient Danish Genomes

We applied the frozen TreeLoom-Episode model to ten ancient Danish query individuals and eight proxy-reference individuals from the imputed genomes analyzed by Irving-Pease et al. (2024). Five earlier individuals date to 5.44–5.67 ka cal BP and five later individuals to 3.97–4.26 ka cal BP. Three fixed 10 Mb windows on chromosomes 20–22 contain 1,158 inferred trees and 8,587 qualified site records.

The model received original phased genotypes and was not fitted again. Person-weighted, span-integrated assignments are 14.27 percentage points higher for EHG and 5.52 points higher for CHG in the later group. The EHG contrast is positive on each chromosome. The pooled CHG contrast depends on chromosome 21, and becomes -0.68 points when that chromosome is omitted. Independent genotype-affinity contrasts using the shared imputation resource do not support the registered directions. These model assignments are descriptive outputs, without local ancestry truth or calibrated mixture fractions. The application illustrates behavior on empirical genealogies rather than confirming a population-history hypothesis.

6 Discussion

TreeLoom provides a direct way to combine reference ancestry, genetic variation and genealogical structure. Its sampled-haplotype interface preserves correspondence along the genome without assuming persistent inferred ancestors. The shared European development comparison is encouraging for this representation. Its biological value must nevertheless be judged by both the amount of correctly assigned sequence and the placement of transitions.

Pipeline	Precision ↑	Recall ↑	F1 ↑
AncestralPaths (C4)	3.95 ± 0.48	$\mathbf{32.36 \pm 2.06}$	7.03 ± 0.78
ARGMix	9.16 ± 1.49	16.41 ± 2.22	11.74 ± 1.73
TreeLoom-Base	10.71 ± 1.21	17.99 ± 2.01	13.42 ± 1.47
TreeLoom-NoMem	$\mathbf{16.31 \pm 1.42}$	22.94 ± 1.58	$\mathbf{19.05 \pm 1.43}$
TreeLoom	15.58 ± 1.08	22.46 ± 2.05	18.38 ± 1.31

Table 4: Present-day boundary metrics (%), mean±simulation SD at 1 kb tolerance. Bold identifies the best mean in each column, including the native comparator's recall and the NoMem control's precision and F1. Matching is positional and does not require the correct source-transition identity.

The matched controls constrain the architectural interpretation. Local structural conditioning is competitive with the examined event interactions, foreign-episode context provides little added accuracy over local conditioning, and richer pair-cell memory does not outperform the simpler controls under the registered criteria. Continued training explains much of the later improvement. The present evidence supports this ancestry framework and its comparative analysis, while the contribution of a distinct cross-tree operator remains an open research question.

The external evaluations also have specific scope. Development validation has been reused, native models retain different training and input choices, and the completed panel is not a verified recreation of the published tables. The Danish application has no ancestry ground truth. A separate tradenetwork diagnostic finds persistence stronger than the tested learned memory models (Supplement), so that experiment does not support a broad generality claim. Independent ancestry tests and completed native benchmarks are the next basis for a stronger conclusion.

7 Conclusion

TreeLoom infers local ancestry from complete inferred genealogies linked by persistent sampled haplotypes. Whole-tree propagation, structural conditioning and genomic recurrence achieve promising development span accuracy. Matched controls show that additional cross-tree memory yields small gains and can reduce boundary quality. These results motivate ancestry models that recover tract structure as well as genomic mass, with final claims tied to independent tests and faithful native comparisons.

Code availability. Source code, experiment configurations and full results are available at https://github.com/alexeyshmelev/ARG-GNN.

References

Baumdicker, F.; Bisschop, G.; Goldstein, D.; et al. 2022. Efficient ancestry and mutation simulation with msprime 1.0. *Genetics*, 220(3).

Bonet, D.; Shanks, C.; Comajoan Cara, M.; Abante, J.; and Ioannidis, A. G. 2026. ARGformer: learning on ancestral recombination graphs with transformers. *Bioinformatics*, 42(Supplement 2): btag438.

Browning, S. R.; Waples, R. K.; and Browning, B. L. 2023. Fast, accurate local ancestry estimation with FLARE. *The American Journal of Human Genetics*, 110(2): 326–335.

Cho, K.; van Merrienboer, B.; Gulcehre, C.; Bahdanau, D.; Bougares, F.; Schwenk, H.; and Bengio, Y. 2014. Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, 1724–1734.

Escrig Molina, J. E.; Chen, B.; and Probst, D. 2026. Graph Set Transformer. arXiv:2606.05116.

Huang, S.; Poursafaei, F.; Danovitch, J.; Fey, M.; Hu, W.; Rossi, E.; Leskovec, J.; Bronstein, M.; Rabusseau, G.; and Rabbany, R. 2023. Temporal Graph Benchmark for Machine Learning on Temporal Graphs. Accepted at NeurIPS 2023 Datasets and Benchmarks Track, arXiv:2307.01026.

Irving-Pease, E. K.; Refoyo-Martínez, A.; Barrie, W.; Ingason, A.; Pearson, A.; Fischer, A.; Sjögren, K.-G.; Halgren, A. S.; Macleod, R.; Demeter, F.; Henriksen, R. A.; Vimala, T.; McColl, H.; Vaughn, A. H.; Speidel, L.; Stern, A. J.; Scorrano, G.; Ramsøe, A.; Schork, A. J.; Rosengren, A.; Zhao, L.; Kristiansen, K.; Iversen, A. K. N.; Fugger, L.; Sudmant, P. H.; Lawson, D. J.; Durbin, R.; Korneliussen, T.; Werge, T.; Allentoft, M. E.; Sikora, M.; Nielsen, R.; Racimo, F.; and Willerslev, E. 2024. The selection landscape and genetic legacy of ancient Eurasians. *Nature*, 625(7994): 312–320.

Li, J.; Wu, R.; Jin, X.; Ma, B.; Chen, L.; and Zheng, Z. 2024. State Space Models on Temporal Graphs: A First-Principles Study. In *Advances in Neural Information Processing Systems*, volume 37, 127030–127058.

Shanks, C.; Bonet, D.; Comajoan Cara, M.; and Ioannidis, A. G. 2026. Graph transformer for ancient ancestry inference. *bioRxiv*. Preprint, version 1, posted 7 April 2026.

Speidel, L.; Forest, M.; Shi, S.; and Myers, S. R. 2019. A method for genome-wide genealogy estimation for thousands of samples. *Nature Genetics*, 51(9): 1321–1329.

Supplementary Material

A Simulation Design and Biological Parameters

The European condition follows the released AncestralPaths formulation associated with Irving-Pease et al. (2024). Table 5 gives its backward-time events. The census at 350 generations defines ancestry supervision before the older source-population mergers. It is separate from the model's observed tree cut.

The panel contains 91 diploids sampled at age zero, together with the released ancient age roster: 81 Bronze Age, seven BAA, 151 Neo and 13 Yam diploids. Pure-reference diploids comprise 51 WHG, 43 EHG, 25 Ana and 14 CHG. This totals 343 query and 133 reference diploids.

Simulations use the Hudson coalescent, discrete genomic coordinates, a uniform recombination rate of 10^{-8} per base per generation and JC69 mutations at 1.25×10^{-8} per base per generation (Baumdicker et al. 2022). The released formulation supplies the demographic events and ancient roster. The 2 Mb sequence length and simulation split define our development condition.

Time (gen.)	Descendant	Ancestral sources
166	Present/Bronze	0.5 Neo, 0.5 Yam
169	BAA	0.25 CHG, 0.75 Ana
177	Yam	0.5 CHG, 0.5 EHG
259	Neo	0.25 WHG, 0.75 Ana
600	WHG, EHG	Shared NE ancestor
800	CHG, Ana	Shared WA ancestor
1,500	NE, WA	Common trunk

Table 5: Executed European demographic events, following the released formulation associated with AncestralPaths. Mixture coefficients are backward-time ancestral proportions. This is a simulation-design table, so no performance winner is applicable.

Population sizes are 50,000 for Present/Bronze and Neo, 20,000 for Yam and BAA, 3,000 for EHG and CHG, 2,000 for WHG, and 5,000 for Ana. The older NE, WA and trunk populations each have size 15,000. Growth and between-event migration rates are zero. The original ancient sampling dates are retained in generations.

Relate uses the observed biallelic variant panel and sample ages. Its initial effective haploid size is 30,000, its mutation rate is 1.25×10^{-8}, and years are converted using 28 years per generation. The European condition includes five population-size estimation iterations and final branch-length re-estimation. All applicable European model adapters consume the same post-estimation tree reconstruction. Mutation-concordance features are computed from observed alleles on these inferred trees.

Research TODO. Before describing any benchmark as a published-paper reproduction, resolve the original ARGMix panel and map configuration and the correspondence between its Post-Neolithic samples and the executed Bronze Age roster. For the American comparison, finalize a source-cited demographic table and focal-sampling protocol alongside the native results.

B Training and Model Hyperparameters

The genetic site encoder has two hidden layers of width 32. It receives donor log-frequency features for the called allele, corresponding availability indicators, the panel frequency of that allele, site entropy and a call indicator. The eight donor groups combine the four reference ancestries with sample ages below or at least 350 generations. Allele-frequency estimates use additive pseudocount 0.5. If any source class represented in a reference age group has no called alleles at a site, all represented classes in that age group are masked for that site. Mean and maximum pooling are combined with call count, missing-pool status and interval span. Leaf features also include the known four-source reference annotation, unknown-label status and sample age. Internal-node features include inferred age, root status and sampled-node status.

Tree hidden states have width 32, residual projections have rank eight, and the bidirectional genomic recurrent network has width 32 per direction. The classifier combines a 32-dimensional local representation and 64-dimensional recurrent history, then maps through width 32 to four ancestry scores. Memory adds 513 parameters and initializes its length scale to 5 kb, constrained to 100 bp–100 kb. Residual output projections start at zero.

The loss is physical-span cross-entropy over all 686 query haplotypes,

$$\mathcal{L} = -\frac{\sum_{tqc} M_{tqc} \log p_{tqc}}{\sum_{tqc} M_{tqc}}, \qquad M_{tqc} = \sum_{i:r(i)=t} m_{iqc}. \tag{13}$$

Here $r(i)$ maps an original scored bin to its native tree interval. Since predictions are constant within an interval, summing its target masses preserves the original objective. A whole simulated chromosome is one optimization example.

Each development stage completes 30 passes over 32 training simulations, or 960 optimizer updates. AdamW uses learning rate 10^{-3}, weight decay 10^{-4}, betas $(0.9, 0.999)$ and epsilon 10^{-8}, with gradient norm clipped at one. Optimizer state is reset for each continuation stage. The reported model family uses training seed 101. Checkpoints every five passes are selected by equal-simulation present-day validation accuracy, followed by lower NLL and then earlier pass.

Table 6 separates capacity and cumulative training. TreeLoom, Pair and Group select added pass 30. TreeLoom-NoMem selects added pass 20, or 640 new updates and 3,520 cumulative updates, while completing the same 960-update run.

The archived dense AncestralPaths classifier uses 5,190,874 of 5,200,000 requested training rows, sampled with replacement, and completes five epochs. All 9,126 missing rows are BAA path-six examples across three simulations: 2.2815% of requested BAA rows and 0.1755% of all requested rows. Other cohort quotas are met. The effect of this shortfall on accuracy is untested. Training loss still decreases at epoch five, so the completed native schedule

Stage	Parameters	New updates	Total
TreeLoom-Base	29,764	960	960
TreeLoom-Clade	30,788	960	1,920
TreeLoom-Local	31,812	960	2,880
TreeLoom / Pair / Group	32,325	960	3,840
TreeLoom-NoMem	31,812	960	3,840

Table 6: Completed training budgets for the lineage used by the memory comparison. Cumulative counts describe final trained models, rather than all selected checkpoints. The final two rows share the TreeLoom-Local initialization. No metric winner is applicable.

Local interaction	Accuracy (%) ↑	NLL ↓
TreeLoom-MLP	96.20 ± 0.70	0.110014
TreeLoom-Tensor	96.27 ± 0.75	0.109332
TreeLoom-Clade	$\mathbf{96.58 \pm 0.60}$	**0.100656**

Table 7: Matched added-parameter and continuation-budget structural comparison. TreeLoom-MLP and TreeLoom-Tensor are alternative clade-event residuals to TreeLoom-Clade. Accuracy is mean±simulation SD. NLL is the equal-simulation mean. Bold identifies the best mean per metric.

does not establish convergence. Training genomic diversity is also different: 32×2 Mb here versus 3×200 Mb in the original study. Its selected model is the final epoch. Current native ARGMix selects epoch nine from 11 completed epochs using its native smoothed, masked, unweighted validation batch-mean loss, with patience two and a 20-epoch cap. The external physical-span scorer does not replace this inner selection rule. These method-specific differences must accompany interpretation of the pipeline comparison.

C Additional Mechanistic and Cohort Results

A matched neighboring-tree-state experiment also has a small paired gain: 0.0153 ± 0.0920 accuracy points, below the 0.10-point practical-gain criterion.

TreeLoom-Episode reaches boundary F1 $18.1558 \pm 1.3790\%$, versus $18.0557 \pm 1.7761\%$ for TreeLoom-Local. The paired difference is 0.1001 ± 0.9632 points. These results concern the completed episode stage, rather than the later memory comparison.

Strict NLL assigns infinite loss when a method gives zero probability to a true source over positive represented mass. AncestralPaths has one such event over 800 query-base pairs, giving infinite mean NLL while retaining finite accuracy. We use its probabilities without flooring or calibration. Sparse-cohort boundary and source-specific metrics can also be undefined when no relevant event occurs. Undefined values are not treated as zero.

D American Baseline Detail

Table 9 gives FLARE's first American source-specific scores. We evaluate original calls using the nearest native marker with midpoint boundaries.

Cohort	TreeLoom-Local ↑	TreeLoom-Episode ↑
Present-day	96.82 ± 0.61	$\mathbf{96.83 \pm 0.59}$
Bronze Age	99.17 ± 0.40	$\mathbf{99.20 \pm 0.32}$
BAA	$\mathbf{99.28 \pm 1.14}$	99.15 ± 1.25
Neo	$\mathbf{99.64 \pm 0.09}$	99.61 ± 0.08
Yam	99.87 ± 0.18	$\mathbf{99.90 \pm 0.13}$
Ancient	$\mathbf{99.49 \pm 0.15}$	99.48 ± 0.13
All queries	$\mathbf{98.7815 \pm 0.2566}$	98.7761 ± 0.2486

Table 8: TreeLoom-Local versus TreeLoom-Episode cohort accuracy (%), mean±simulation SD. Bold identifies the larger mean. The final row retains additional mean precision to distinguish the close ranking.

Source	Precision	Recall	F1	True records
EUR	97.32	97.68	97.50	3,274
EAS	98.53	98.64	98.58	5,287
AFR	97.68	96.46	97.06	1,439

Table 9: FLARE on one American validation simulation. Metrics are percentages. Rows describe different source classes rather than competing methods, so no between-method winner is marked. There is no between-simulation SD.

Research TODO. Complete the native ARGformer pretraining, contrastive learning and retrieval stages. Its paper specifies temperature 0.05 and inverse-frequency anchor weighting. The released contrastive route defaults to temperature 0.30 and can fall back to uniform weights when its training loader is unavailable. The completed benchmark must identify the executed condition and resolve these differences before it is described as reproducing the paper.

E Initial Cross-Domain Diagnostic

The trade-network task from Temporal Graph Benchmark (Huang et al. 2023) provides an initial transfer diagnostic. Models predict before observing the current year's edges. Validation uses 2010–2013 and test uses 2014–2016. All three learned models share a residual persistence prior and complete 100 epochs with a fixed seed. Test predictions precede target access. All learned models are below persistence (Table 10).

Method	Validation ↑	Test ↑
Persistence	$\mathbf{0.8672 \pm 0.0107}$	$\mathbf{0.8469 \pm 0.0144}$
First-order memory	0.8574 ± 0.0073	0.8324 ± 0.0113
Conditional triple	0.8567 ± 0.0054	0.8305 ± 0.0110
Tree only	0.8581 ± 0.0056	0.8271 ± 0.0113
Seven-year EMA	0.8429 ± 0.0085	0.8090 ± 0.0049

Table 10: Trade-network NDCG@10, mean±year SD. Years are serially dependent, and SD is not training-seed uncertainty. Bold identifies the best mean. This causal diagnostic differs from published leaderboard settings.